

SAMUEL P. VARA

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EDUCATION

University of Pennsylvania, Philadelphia

Aug 2024-May 2026

Masters in Computer and Information Science (with conc. Artificial Intelligence)

Coursework: Wharton Product Design, Applied Machine Learning, Natural Language Processing, Networked Systems, Software Systems

SRM Institute of Science and Technology, India

Jun 2020-Jun 2024

Bachelor of Technology in Computer Science and Engineering

Coursework: Computer Architecture, Operating Systems, Neuro and Fuzzy Logic, Data Mining, Software Systems

SKILLS

Languages	Python (Django, Flask), Java, SQL (MySQL, PostgreSQL), NoSQL (MongoDB), C++, C
UI Design	Angular 17, React, Node, Figma, Javascript, HTML, CSS, Vercel, Render
Tools	BeautifulSoup, Sci-kit learn, TensorFlow, PyTorch, Keras, Cuda, Pandas, Git, CellPose, Docker, Kubernetes, Helm
Algorithms	Computer Vision (CNN, YOLO), Natural Language Processing (SVM, TF-IDF, LDF, Transformers-GPT, LSTM), CI/CD
Cloud	AWS (EC2, EKS, Lambda, ElastiCache, S3, SageMaker), Google Cloud Platform (GCP), Product Experience Management

EXPERIENCE

Chief Executive Officer | Ascend AI Innovations - Philadelphia, PA

Jul 2026-Present

- Lead engineering and product for a portfolio of multitenant AI agent platforms serving clients across 65+ industries.
- Architected the shared stack: React and Vite frontends on Vercel, Cloudflare Workers with KV for state, Anthropic Claude for inference, driven by a configuration layer that spins up a new client platform from shared components.
- Built BoostUrBusiness Intelligence with specialised agents with persistent per agent context in KV, live CRM sync, and automated publishing to LinkedIn, Instagram and Facebook with generative image assets.

Lead Software Developer | Monell Chemical Senses Center – Philadelphia, PA

Jan 2025-Present

- Saved 90h/week in annotation and classification times of Nodose neurons by developing custom Machine Learning model [SoFiA].
- Increased productivity by 60% of Dr. Guillaume de Lartigue's laboratory by developing iSAM, a neuron-clustering platform that analyzes activity patterns to identify functional subpopulations.

Research Assistant | The Wharton School, University of Pennsylvania - Philadelphia, PA

Jul 2025-May 2026

- Increased marketing efficiency by 80% by integrating AI with Google SERP to accelerate global author mapping [Project ATLAS].
- Engineered CompoApp, an AI-driven product ideation and cost-prediction system that integrates market analysis with AAPMR medical pricing data, leveraging custom LLM pipelines to forecast billing costs and product feasibility.
- Led and developed a fast indexing algorithm to discover Wharton's resources using custom Transformer model [Project Andrea].

Head Teaching Assistant | University of Pennsylvania - Philadelphia, PA

Jul 2024-Dec 2024

- Supported instruction on foundational conversational AI theories and principles for CIS 7000 under Prof. Lyle Ungar.
- Mentored 80 students to build conversational agents using bidirectional Encoder-Decoder architecture.
- Designed advanced computational conversational strategies leveraging NLP techniques to improve AI chatbot performance and dialogue quality.

Data Analytics and Deep Learning Intern | Amazon Web Services and National University of Singapore

Jun 2023-Jul 2023

- Engineered a state-of-the-art oral cancer detection model using a novel Composite Convolutional Neural Network (CCNN), achieving over 95% diagnostic accuracy, exceeding industry benchmarks, and winning the 2023 Best Project Award.
- Partnered with ComSci professors to design and deploy advanced data analysis models, leveraging AWS SageMaker for scalable machine learning pipelines.

Founder and Lead Research Scientist | SRM Visual Computing Research Community - India

Jan 2022-Jun 2024

- Collaborated with cross-functional teams to translate visual research insights into product and marketing strategies, delivering high-impact solutions and successfully selling multiple products to clients nationwide.
- Led the design and execution of 80+ large-scale projects across emerging technologies, while mentoring and managing 500+ students, setting technical direction, and cultivating a high-performance culture of innovation, accountability, and engineering excellence.

PROJECTS

Sofia, Neuron Detection and Firing Pattern Detections in Nodose Ganglion | Deep Learning, CellPose, Computer Vision

HealthSphere: AI Integrated Disease Detection and Communication | Reinforcement Learning, Deep Learning, CCNN, NLP

AI-Powered Project Atelier Platform for Research Mapping | OpenAI API (GPT-4), Google SerpAPI, Pandas, LangChain, Streamlit

Andrea, Wharton- Penn AI Assistant | OpenAI API, Bert, SerpAPI, AWS, Railway, PostgreSQL, StreamLit

HONORS & AWARDS

Honorable Mention for Outstanding Research Award 2026

Winner of Perelman School of Medicine - Immune Health Hackathon 2025

Academic Excellence Award and Special Recognition Award - SRM University (2024)